

# ARAVIND RAM

## SITE RELIABILITY ENGINEER

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## PROFFESIONAL SUMMARY

Cloud Native engineer with 2+ years of hands-on Kubernetes platform engineering at scale. Designed and operated EKS clusters running 200+ nodes and 120 TB+ data workloads. Expert in Helm chart authoring, ArgoCD GitOps, custom Kubernetes operators, and production-grade cluster hardening with OPA Gatekeeper.

## SKILLS

- SRE & Operations: Incident Management, On-Call, SLO/SLI/Error Budgets, Runbook Automation, Capacity Planning, Chaos Engineering
- Kubernetes: EKS, Karpenter, Helm, ArgoCD, Kustomize, Gatekeeper, ExternalDNS, ALB Controller, ClickHouse Operator
- Observability: Prometheus, Grafana, AlertManager, PromQL, Elastic Stack, Kibana, Haystack, Node Exporter
- IaC & Cloud: Terraform, AWS (EKS, EC2, VPC, IAM, S3, Route53, ACM, ASG), GitOps
- Languages & Tools: Python, Bash, Jenkins, GitHub Actions, Docker, Linux Administration

## EXPERIENCE

### FRESHWORKS, CHENNAI

2023- PRESENT

#### Site Reliability Engineer

- Kubernetes Platform Operations
  - Provisioned and operated production EKS clusters on AWS with Terraform (VPC, IAM, Route53, ACM, S3, ASG, Security Groups) for inhouse Clickhouse (Analytics Workload) and MongoDB Clusters across US, EUC, IND, AU and EUN Regions.
  - Configured ArgoCD GitOps pipelines; deployed Karpenter, Gatekeeper, ExternalDNS, ALB Controller, ClickHouse Operator
  - Migrated entire deployment framework from Kustomize to Helm; improved maintainability and reduced deployment variance
  - Optimized ClickHouse SELECT query performance by ~40% through Lazy Materialization evaluation
- Automation & CI/CD
  - Built GitHub Actions workflows for Helm validation, ECR image sync, automated release tagging, ArgoCD deployment updates
  - Developed Python-based tenant provisioning: dynamic manifest generation for ClickHouse, ZooKeeper, and Envoy; ArgoCD integration eliminated manual onboarding
- Observability
  - Deployed full observability stack: Prometheus, Grafana dashboards, AlertManager rules (PromQL), Node Exporter, Elastic + Kibana for centralized log aggregation
  - Integrated Slack alerting for proactive incident detection; reduced MTTD significantly
  - Built AWS Cost Explorer + Lambda pipeline for per-tenant infrastructure cost visibility

- Reliability & Incident Response

- Been on-call rotation for ClickHouse platform serving 120 TB of analytics data across 200+ servers and MongoDB Platform of 80TB with around 60+ Servers.
- Reduced CPU throttling by ~90% by implementing Kubernetes CPU Manager static policy for ClickHouse workloads
- Built SOPCTL incident assistant integrating Slack alerts → Temporal workflows → automated SOP execution → LLM-generated RCA, accelerating incident resolution
- Designed DR framework: hourly ClickHouse backups, S3 cross-region replication, automated restore workflows; improved RTO from hours to minutes

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## PROJECTS

- ClickHouse MCP Server — Built AI-assisted operational tooling (FastMCP, MetaMCP) for query pattern analysis, Prometheus alert correlation, and optimization recommendations
- MongoDB Platform Migration — Led PSMDB → in-house CSMDDB operator migration covering sharding, backup, PITR, lifecycle management; built Jenkins CI pipelines
- DocumentDB → MongoDB — Migrated 16 TB of production data; validated compatibility and data integrity throughout

## EDUCATION

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Bachelor of Science (BS) Degree in Data Science and Applications

Indian Institute of technology, Madras 2023 - 2027

## COURSEWORK

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CKAD: Certified Kubernetes Application Developer

## LANGUAGES

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Tamil - Native

English - Fluent